XAI: Intrinsically explainable models

Responsible AI course, 5.11.2026

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Learning goals

By the end of this lecture, you should:

- Be able to define and use the ProtoPNet model for obtaining explanations through learned prototypes
- Be able to define and use concept bottleneck models (CBMs)
- Be familiar with criticism against CBMs, most notably leakage and majority voting

Last time

Last time, we talked about various types of saliency map



Figure: GradCAM saliency map for a cat species prediction

Gradients

- For a function $f : \mathbb{R}^d \to \mathbb{R}$, the gradient

$$\nabla_x f = (\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_d})$$

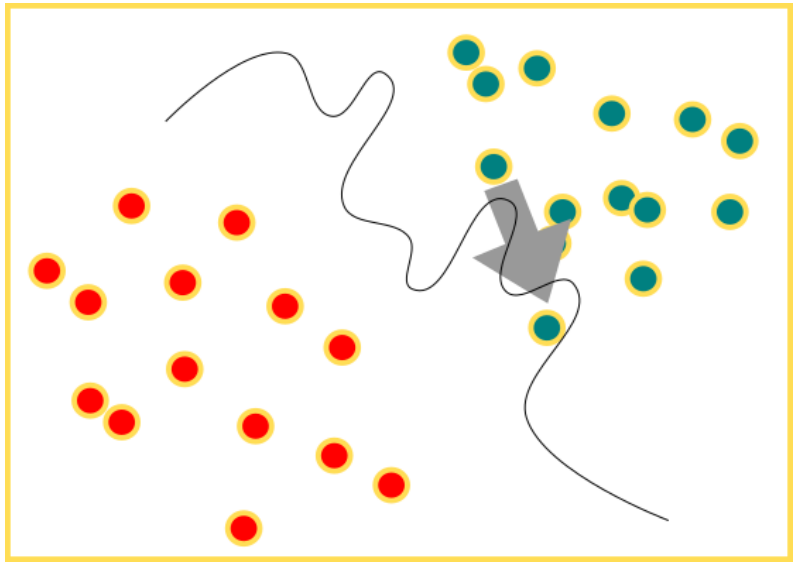
  at $x_0 \in \mathbb{R}^d$ is a vector in $\mathbb{R}^d$ pointing in the direction of maximal slope.
- For an image classifier $f : \mathbb{R}^{H \times W} \to \mathbb{R}$ predicting probability of positive class, and an image $x \in \mathbb{R}^{H \times W}$, what is ∇f?

Gradients

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- For an image classifier $f : \mathbb{R}^{H \times W} \to \mathbb{R}$ predicting probability of positive class, and an image $x \in \mathbb{R}^{H \times W}$, what is ∇f?
- A matrix (image) in $\mathbb{R}^{H \times W}$, pointing in the direction of the positive class.

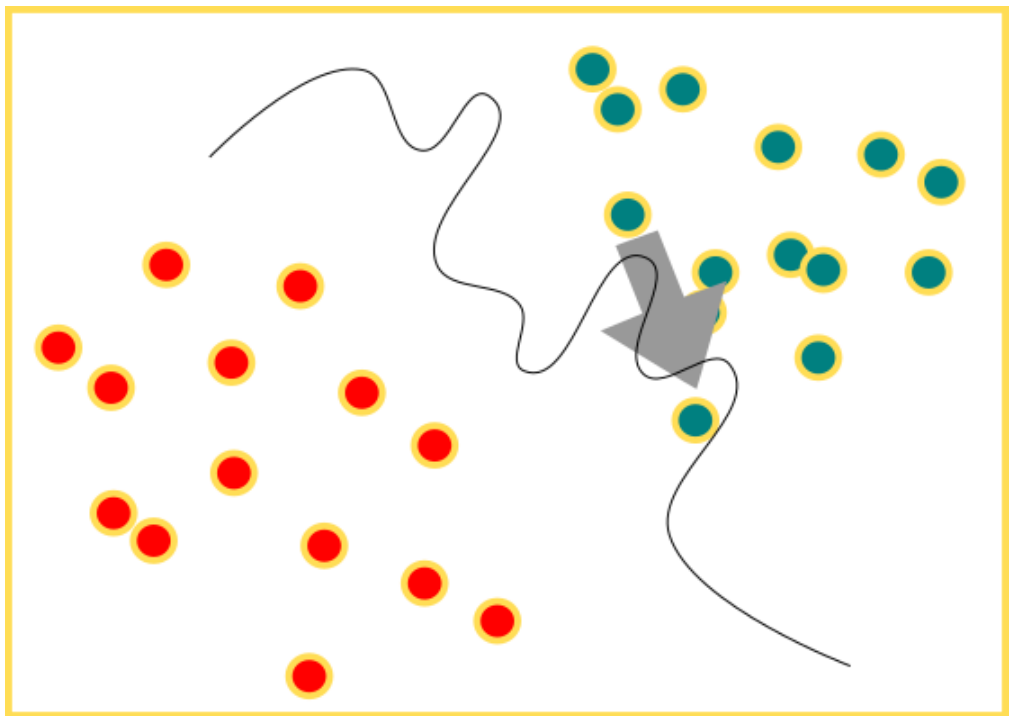


Vanilla saliency maps: Gradients

The vanilla gradient saliency map for the image is simply given by the gradient

$$M_c = \nabla_x f_c(\theta, x)$$

of the (usually the correct) c^{th} class prediction of the network f at x.



GradCAM

- Post-hoc method
- Activation mapping weights computed via gradients:

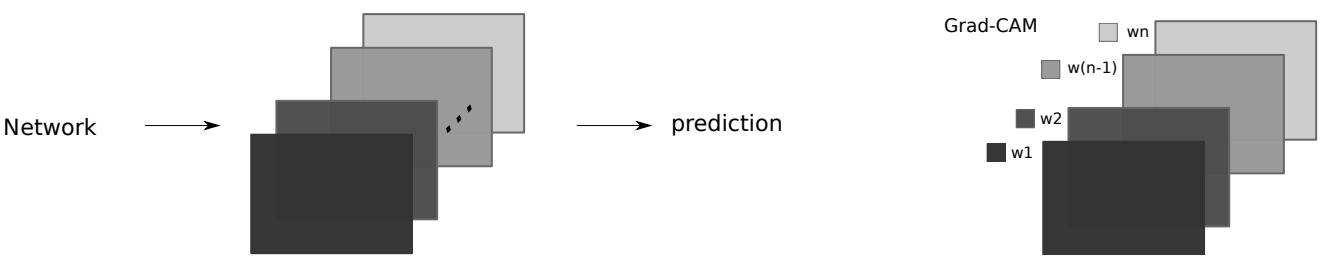
$$w_k^c = \text{mean}(\nabla_{A^k} f_c(x))$$

where $f_l(x)$ is the (before softmax) prediction of the label c for input image x, A^k is the features of the kth penultimate channel.

- Saliency map:

$$M_c(x) = ReLU(\sum_k w_k^c A^k)$$

- ReLU turns off negative contributions – more intuitive output



Alternative: Image-retrieval / nearest neighbor based prototypical XAI

Deep kNN for interpretable AI

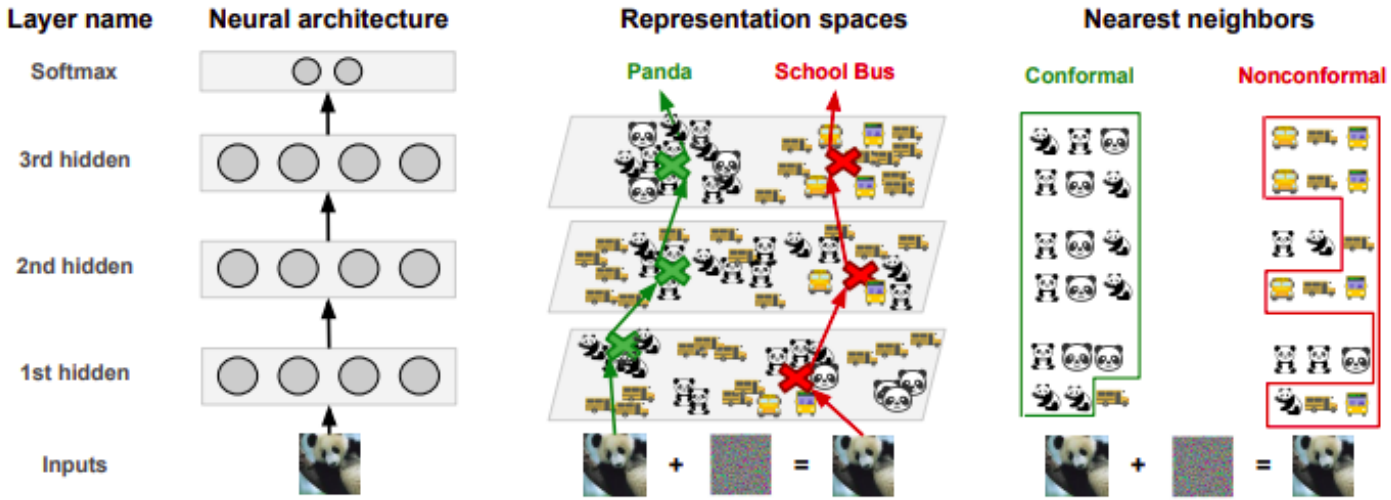


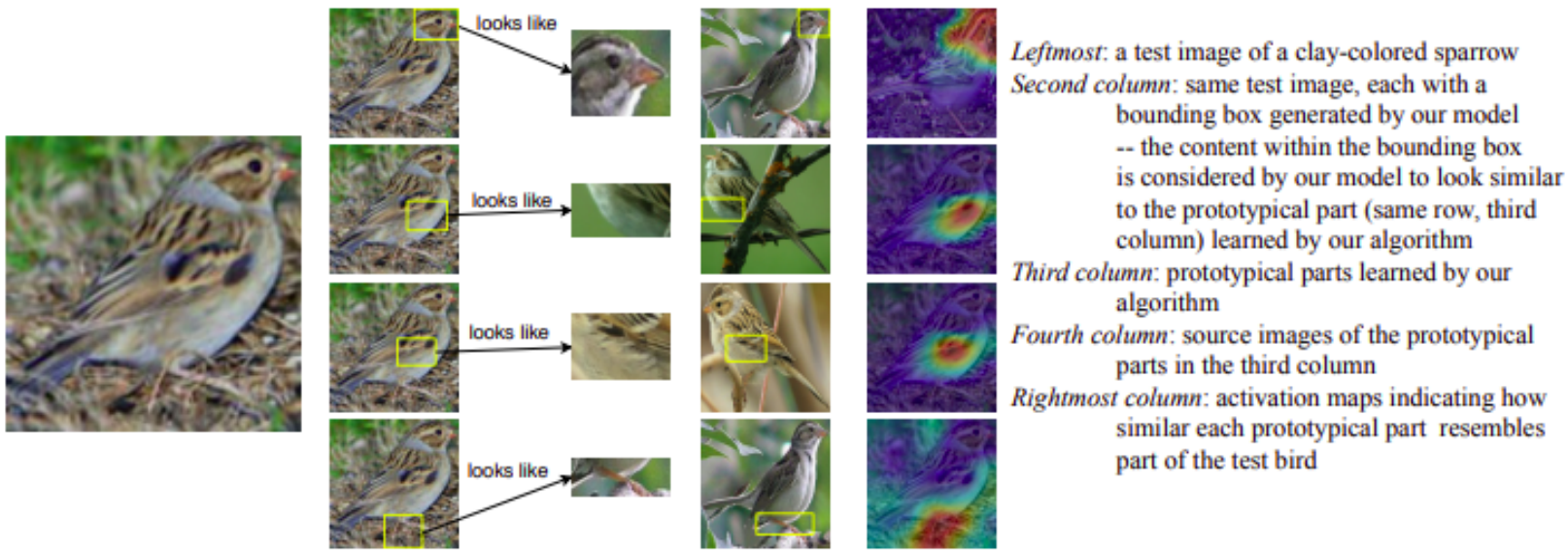
Figure: Papernot and McDaniel, Deep k-Nearest Neighbors: Towards Confident, Interpretable and Robust Deep, Learning, 2018

ProtoPNet

Chen et al, **This Looks Like That: Deep Learning for Interpretable Image Recognition**, NeurIPS 2019

ProtoPNet

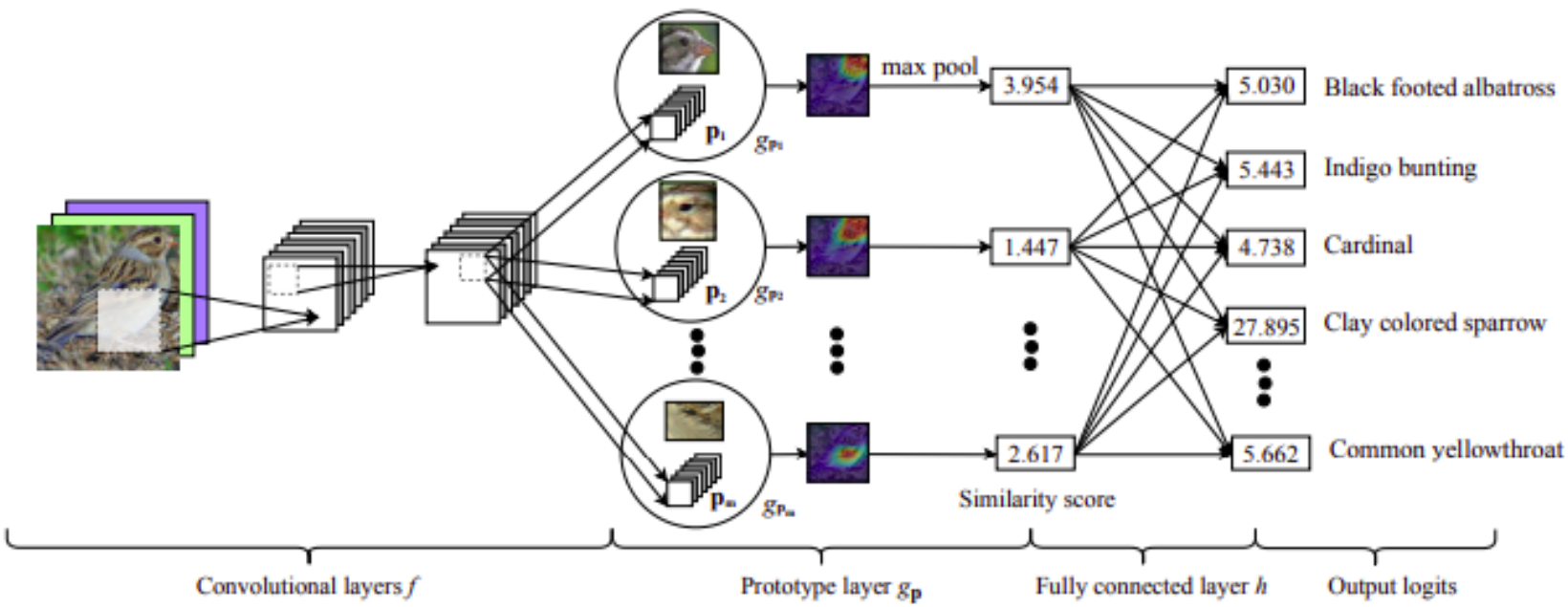
Motivation: Classification decisions made by prototyping particular elements (patches) from the image



Leftmost: a test image of a clay-colored sparrow
Second column: same test image, each with a bounding box generated by our model -- the content within the bounding box is considered by our model to look similar to the prototypical part (same row, third column) learned by our algorithm
Third column: prototypical parts learned by our algorithm
Fourth column: source images of the prototypical parts in the third column
Rightmost column: activation maps indicating how similar each prototypical part resembles part of the test bird

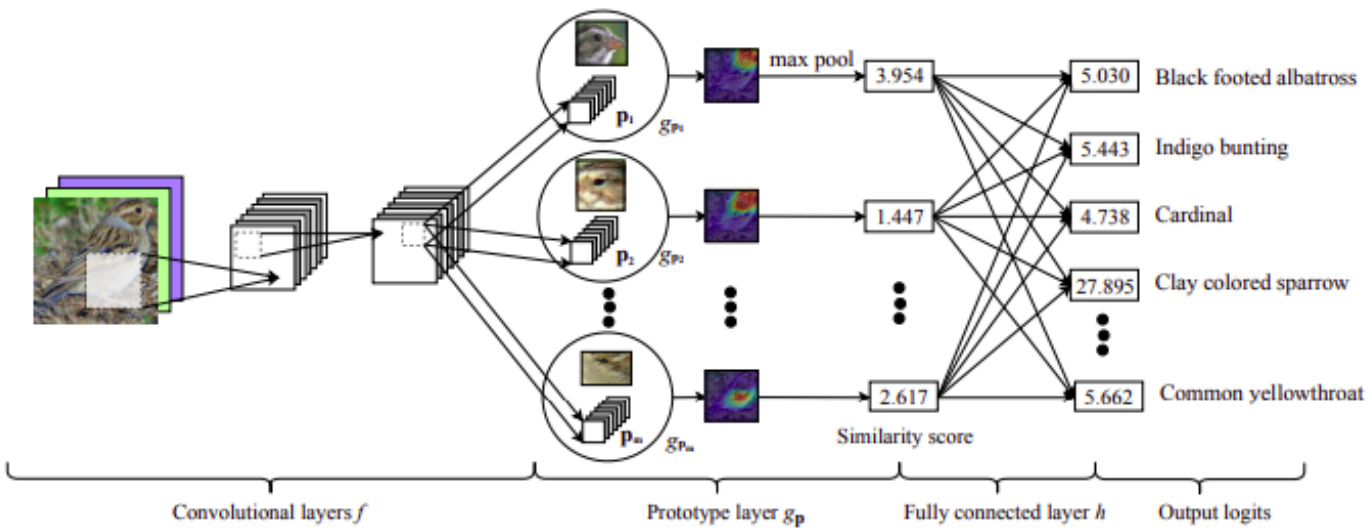
ProtoPNet

Architecture overview: Perform classification using similarity to learned prototypes



Inferring and Using Prototype similarities

- Each $1 \times 1 \times D$ "pixel" in a latent image $\mathbf{z}$ of size $H \times W \times D$ from the prototype layer, corresponds to a patch in the original image (its receptive field)
- Prototypes – we'll see how to learn them later – are also $1 \times 1 \times D$ pixels corresponding to input patches
- For each prototype patch, its similarity to each patch from $\mathbf{z}$ is used to obtain a *prototype activation map* over $\mathbf{z}$



Learning prototypes

- A fixed number (10) of prototypes per class

$$\min_{\mathbf{P}, w_{\text{conv}}} \frac{1}{n} \sum_{i=1}^{n} \text{CrsEnt}(h \circ g_{\mathbf{p}} \circ f(\mathbf{x_i}), \mathbf{y_i}) + \lambda_1 \text{Clst} + \lambda_2 \text{Sep}, \quad \text{where Clst and Sep are defined by}$$

$$\text{Clst} = \frac{1}{n} \sum_{i=1}^{n} \min_{j:\mathbf{p}_j \in \mathbf{P}_{y_i}} \min_{\mathbf{z} \in \text{patches}(f(\mathbf{x}_i))} \|\mathbf{z} - \mathbf{p}_j\|_2^2; \text{Sep} = -\frac{1}{n} \sum_{i=1}^{n} \min_{j:\mathbf{p}_j \notin \mathbf{P}_{y_i}} \min_{\mathbf{z} \in \text{patches}(f(\mathbf{x}_i))} \|\mathbf{z} - \mathbf{p}_j\|_2^2.$$

- Optimization scheme to encourage correct prediction while also keeping i) every image having some patch nearby a prototype of its class (Clst) while keeping prototypes from other classes further away (Sep)
- Trained cyclically: First weights of f and prototypes; then projecting prototypes, then the final FCN.

Reasoning process



Related model: Penalizing use of confounding information

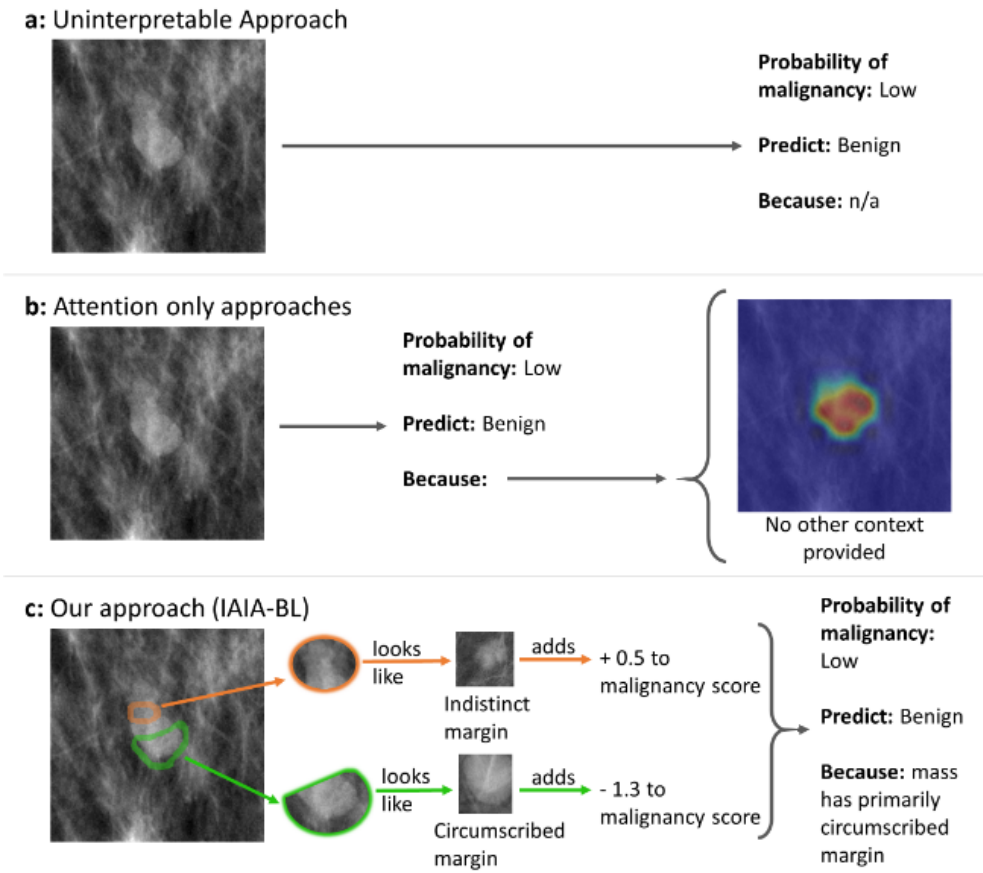
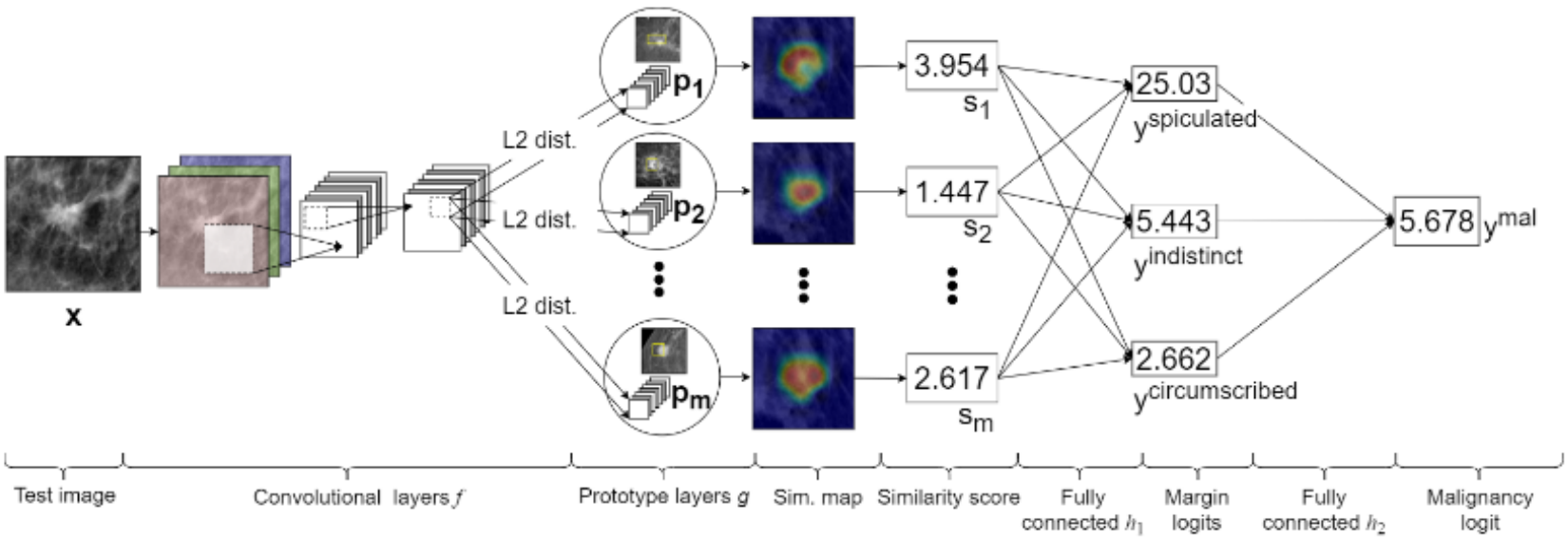
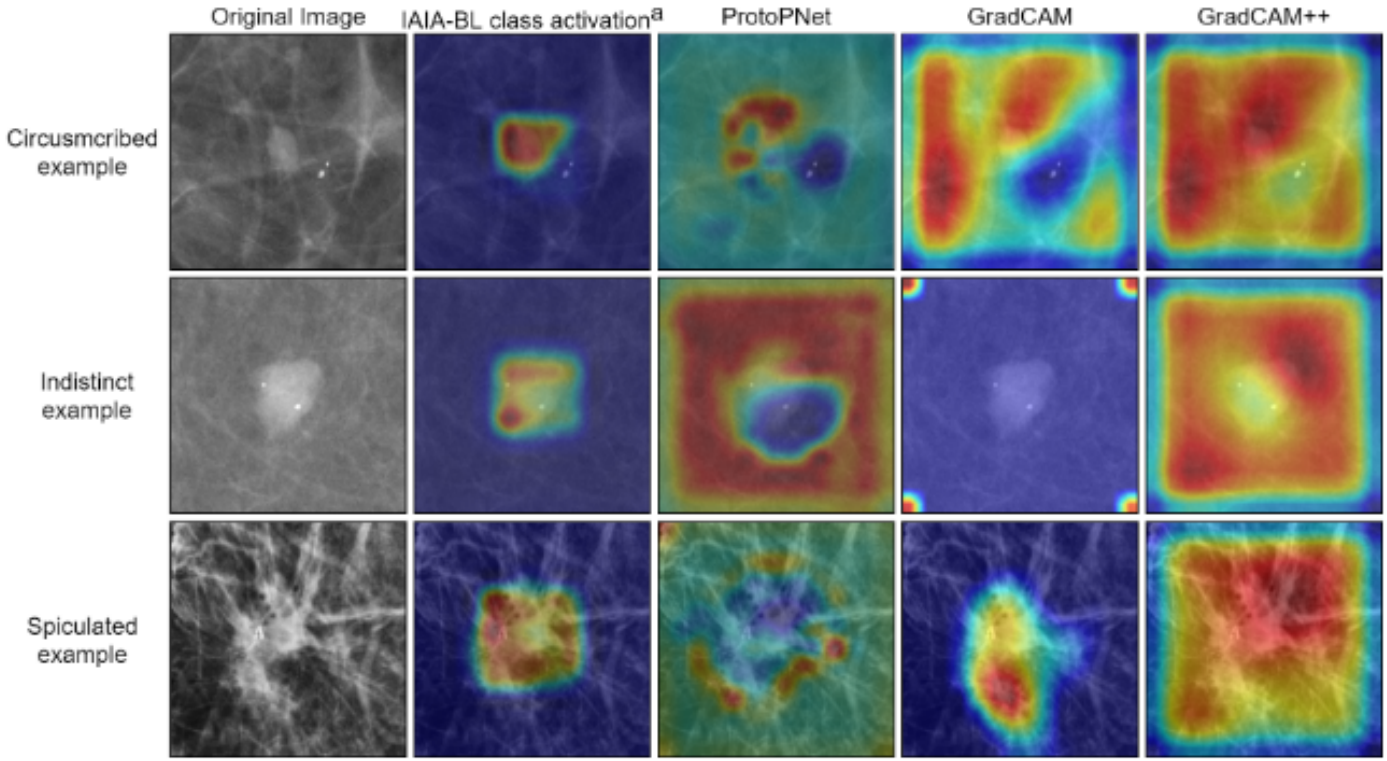


Figure: Barnett et al, A case-based interpretable deep learning model for classification of mass lesions in digital mammography, Nature Machine Intelligence 2021.

Modified Architecture



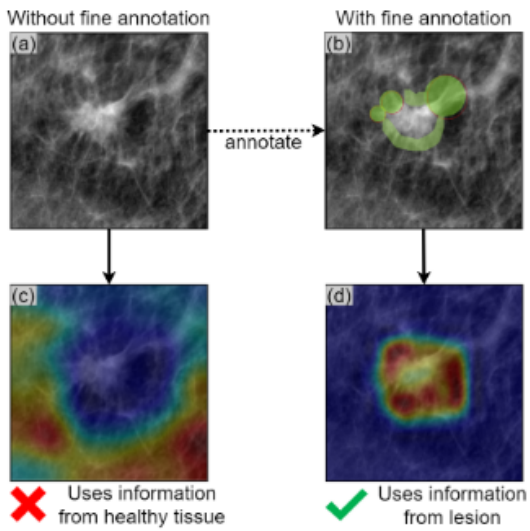
Example results



Modified Loss

$$\text{FineLoss} = \sum_{i \in D'} \left(\sum_{j:\text{class}(\mathbf{p}_j)=y_i^{\text{margin}}} \|\mathbf{m}_i \odot \text{Upsample}(g_{\mathbf{p}_j}(f(\mathbf{x}_i)))\|_2 + \sum_{j:\text{class}(\mathbf{p}_j)\neq y_i^{\text{margin}}} \|g_{\mathbf{p}_j}(f(\mathbf{x}_i))\|_2 \right)$$

Discussion: Is this a good idea? Do you see any dangers?



Summary: ProtoPNet

- **Pros:**
- **Cons:**

Summary: ProtoPNet

- **Pros:** Built in explainability in the model
  Discuss: But is that completely true?
- **Cons:** Does not generalize to generic problems. Only works
  - Where you know what information you expect the algorithm to use
  - Where the relevant information is spatially restricted
  - Where potential confounding information does not overlap with relevant information

Intrinsic explainability: A polarized debate

nature machine intelligence

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nature > nature machine intelligence > perspectives > article

Perspective | Published: 13 May 2019

Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead

Cynthia Rudin

Nature Machine Intelligence **1**, 206–215 (2019) | Cite this article

72k Accesses | **2802** Citations | **496** Altmetric | Metrics

A preprint version of the article is available at arXiv.

Abstract

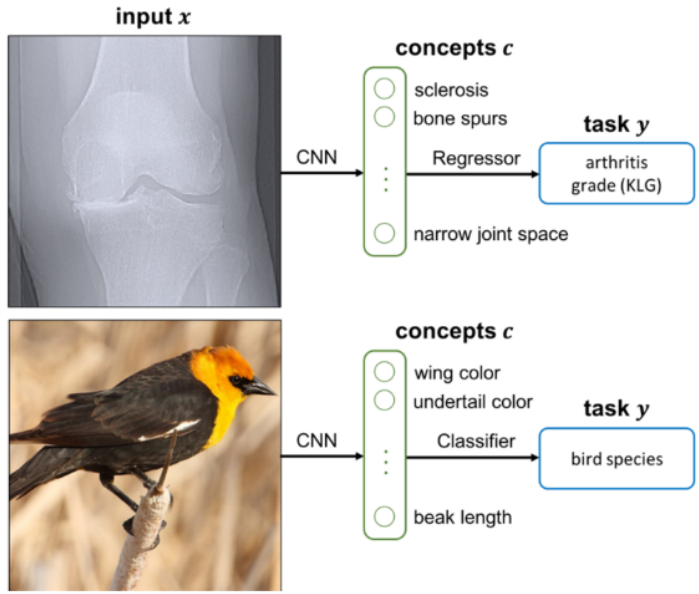
Black box machine learning models are currently being used for high-stakes decision making throughout society, causing problems in healthcare, criminal justice and other domains.

Concept bottleneck models

Koh et al, Concept Bottleneck Models, ICML 2020

Concept bottleneck models

Motivation: Explanatory pre-defined concepts that you can interact with



- Training points (x_i, c_i, y_i) for input x_i, concept vectors c_i, outputs y_i
- Concept bottleneck $\hat{y} = f(g(x))$, where $\hat{c} = g(x)$ estimates concepts, and $y = f(c)$ estimates outputs from concepts

Training concept bottleneck models

- Task performance: in predicting y
- Concept performance: in predicting c

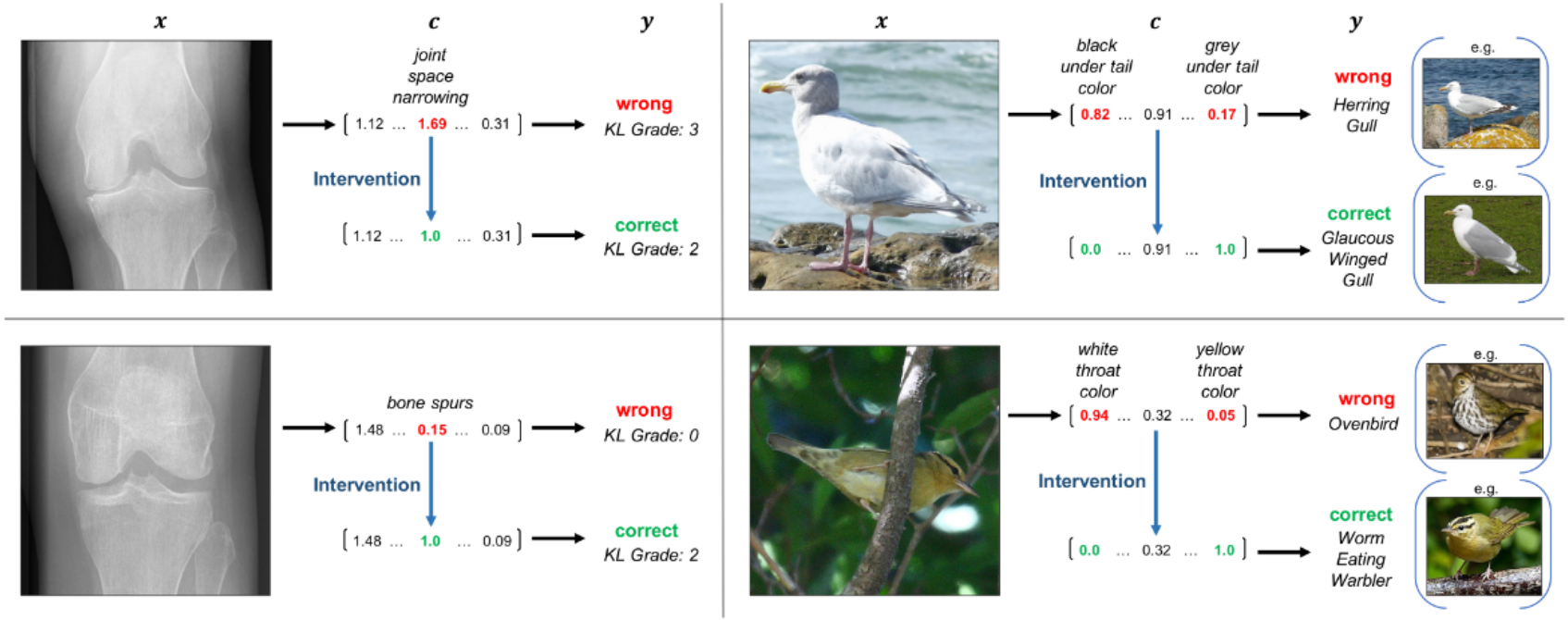
1. The *independent bottleneck* learns $\hat{f}$ and $\hat{g}$ independently: $\hat{f} = \arg\min_f \sum_i L_Y(f(c^{(i)}); y^{(i)})$, and $\hat{g} = \arg\min_g \sum_{i,j} L_{C_j}(g_j(x^{(i)}); c_j^{(i)})$. While $\hat{f}$ is trained using the true c, at test time it still takes $\hat{g}(x)$ as input.

2. The *sequential bottleneck* first learns $\hat{g}$ in the same way as above. It then uses the concept predictions $\hat{g}(x)$ to learn $\hat{f} = \arg\min_f \sum_i L_Y(f(\hat{g}(x^{(i)})); y^{(i)})$.

3. The *joint bottleneck* minimizes the weighted sum $\hat{f}, \hat{g} = \arg\min_{f,g} \sum_i \left[L_Y(f(g(x^{(i)})); y^{(i)}) + \sum_j \lambda L_{C_j}(g(x^{(i)}); c^{(i)}) \right]$ for some $\lambda > 0$.

4. The *standard model* ignores concepts and directly minimizes $\hat{f}, \hat{g} = \arg\min_{f,g} \sum_i L_Y(f(g(x^{(i)})); y^{(i)})$.

Intervention

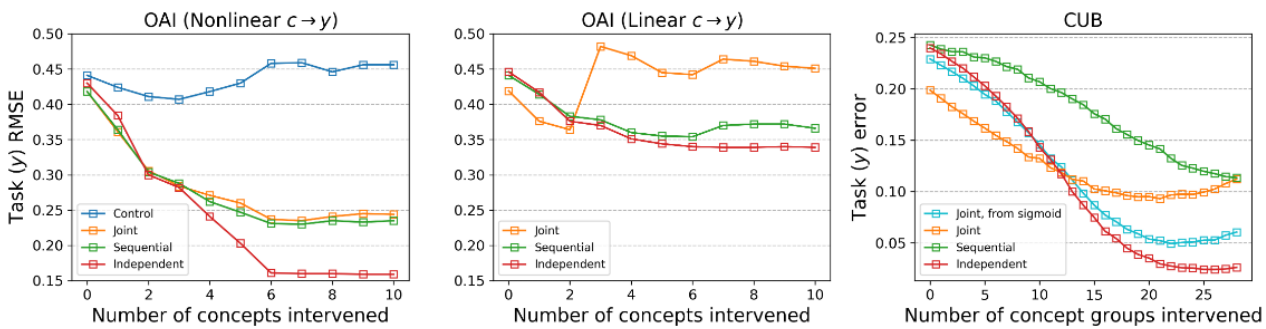


Concept bottleneck models – what are they good for?

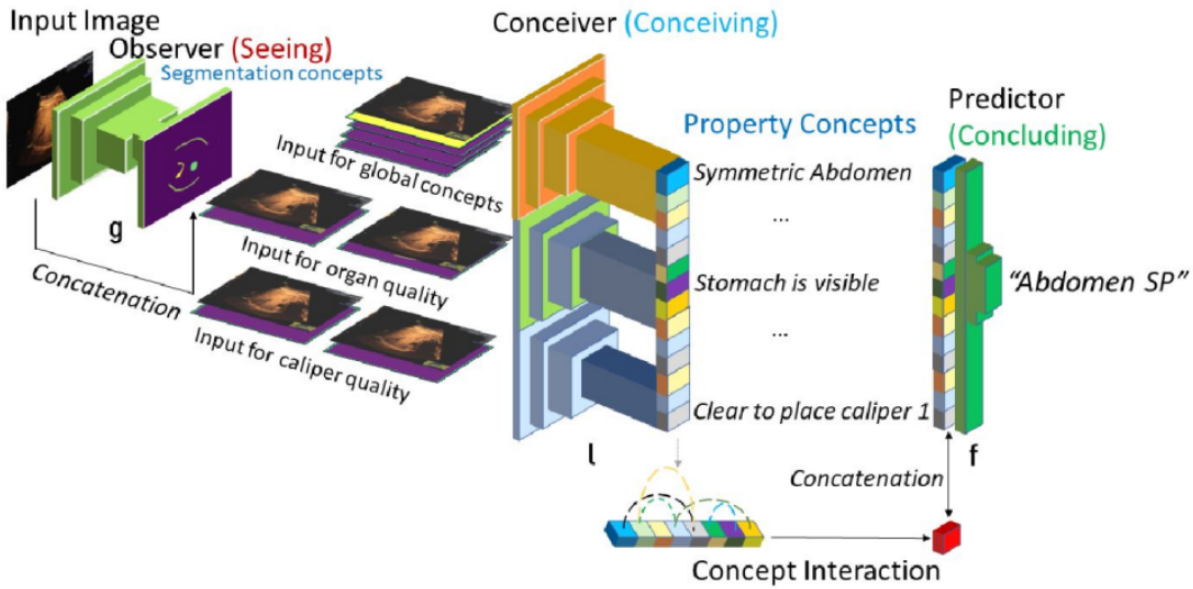
- (Potentially) improving performance of the model

Model	y RMSE (OAI)	y Error (CUB)
Independent	0.435± 0.024	0.240±0.012
Sequential	0.418± 0.004	0.243±0.006
Joint	0.418± 0.004	0.199±0.006
Standard	0.441± 0.006	0.175±0.008
No bottleneck	0.443 ± 0.008	0.173±0.003
Multitask	0.425± 0.010	0.162±0.002

- Allowing intervention with the concepts, further improving performance



Progressive CBMs for fetal ultrasound quality assessment

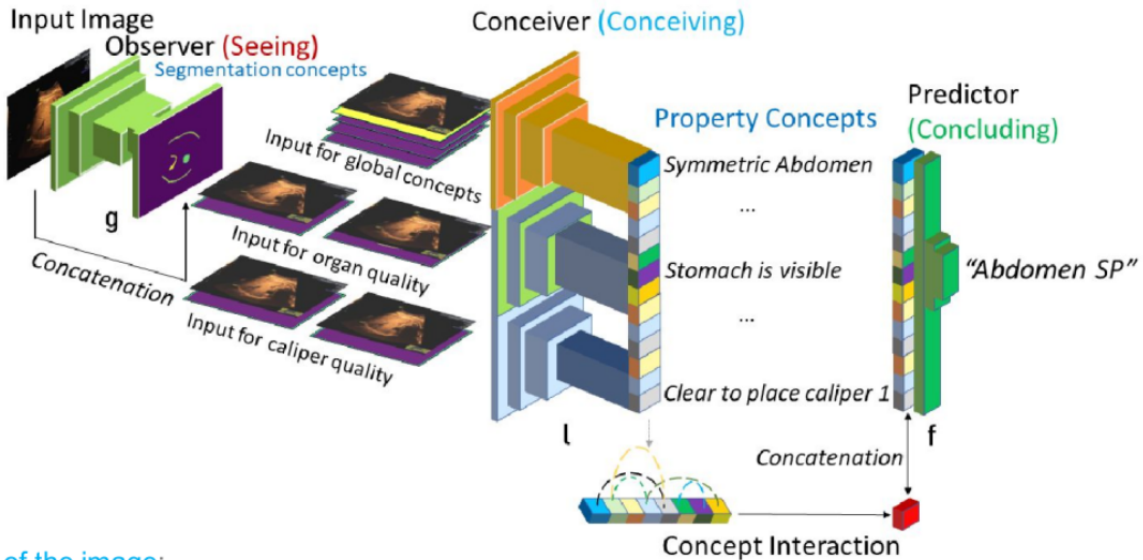


Progressive **C**oncept **B**ottleneck **M**odels (PCBMs)

SCAN ME

Lin et al., I saw, I conceived, I concluded: Progressive Concepts as Bottlenecks. arXiv preprint arXiv: 2211.10630

Progressive CBMs for fetal ultrasound quality assessment



ISUOG
- Symmetrical plane;
- Plane showing stomach bubble;
- Plane showing portal sinus;
- Kidneys not visible;
- Abdomen occupying more than half of the image;

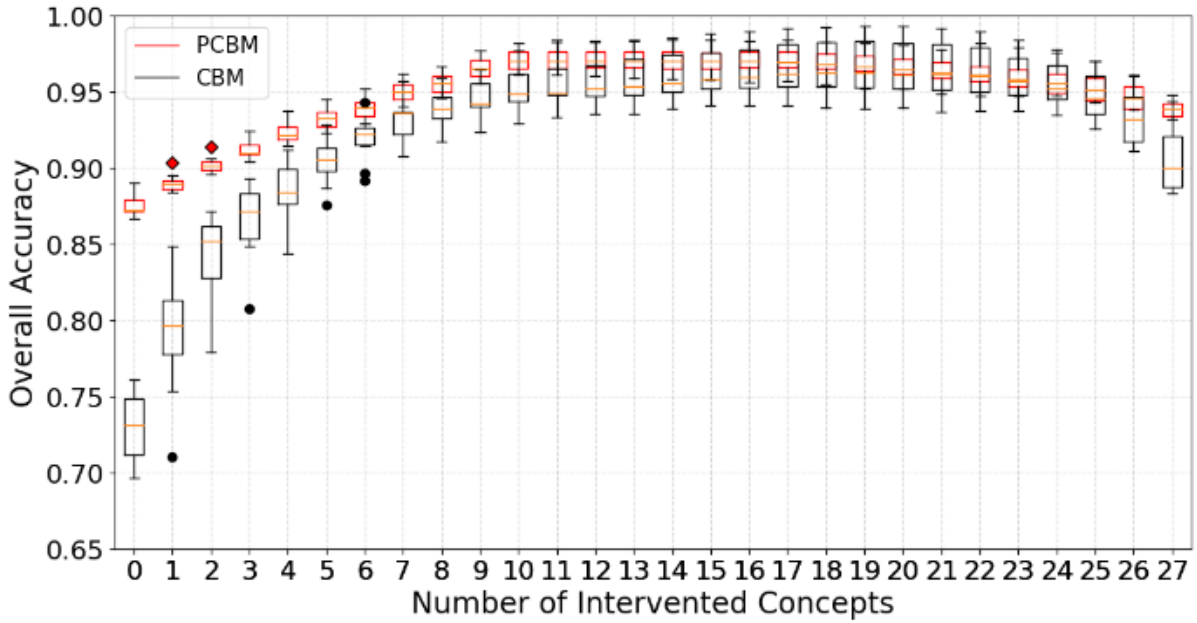
*Since there are **clear stomach bubble** **and** **portal sinus** in the image, **the scan is symmetric**, **kidney not visible**, and **abdomen occupies more than half of the image**, this is an acceptable ultrasound scan of Abdomen (**abdomen standard plane**).*

Progressive CBMs for fetal ultrasound quality assessment

Method	OA (%)	MA (%)	MCC (%)
Standard Model	62.71 ± 5.67	61.64 ± 1.68	57.62 ± 5.12
SonoNet-32 [5]	68.83 ± 3.13	62.73 ± 0.81	63.81 ± 3.02
SASceneNet-18 [31]	78.89 ± 3.30	78.46 ± 1.00	75.71 ± 3.32
CBM [26]	72.24 ± 1.99	72.22 ± 0.45	68.24 ± 1.92
PCBM (ours)	$\mathbf{87.53} \pm 0.66$	$\mathbf{88.84} \pm 0.86$	$\mathbf{85.63} \pm 0.78$

Table 2. Classification performance of different models on the ultrasound dataset. 'OA' means overall classification accuracy over instances; 'MA' refers to the average classification accuracy over categories; 'MCC' stands for Matthews correlation coefficient.

Progressive CBMs for fetal ultrasound quality assessment



Progressive CBMs for fetal ultrasound quality assessment

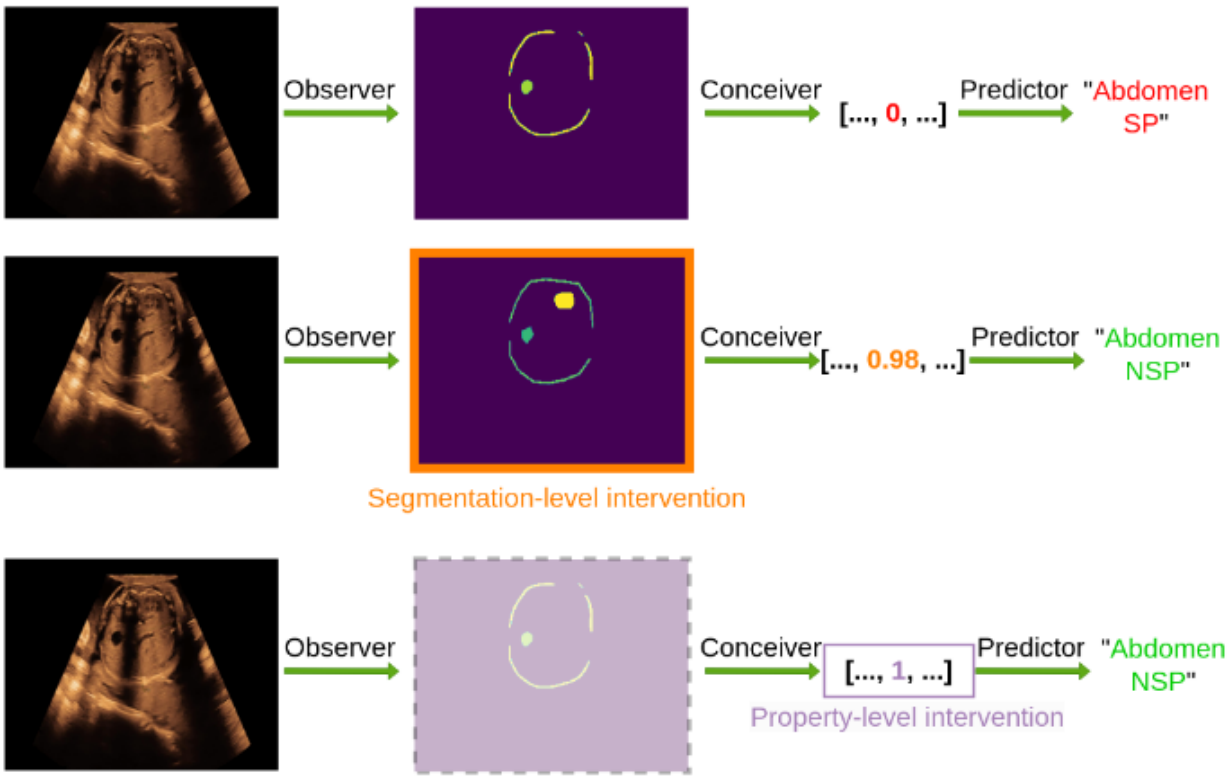
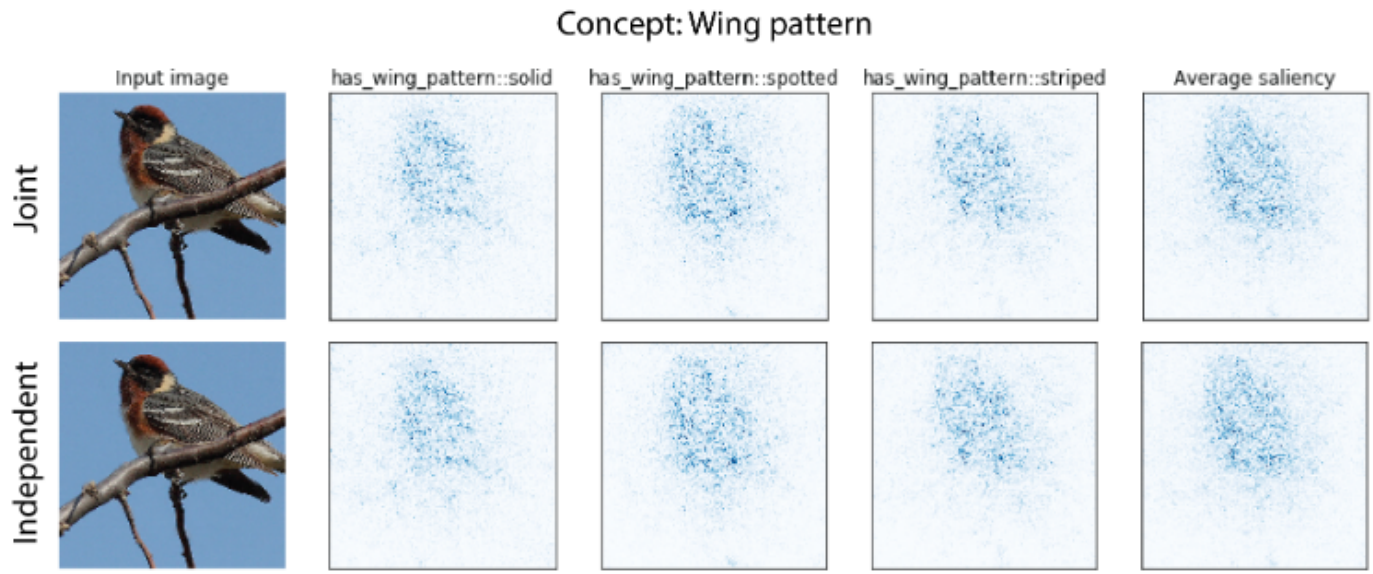


Figure 11. An example of successful human intervention at different levels is the proposed PCBM. The top row is a wrong prediction from our model because the model does not recognize the kidney in the image. The middle and the bottom row illustrates the human intervention at the segmentation and property level respectively. The number in the figure stands for the predicted kidney quality concept.

Arguments against concept bottleneck models

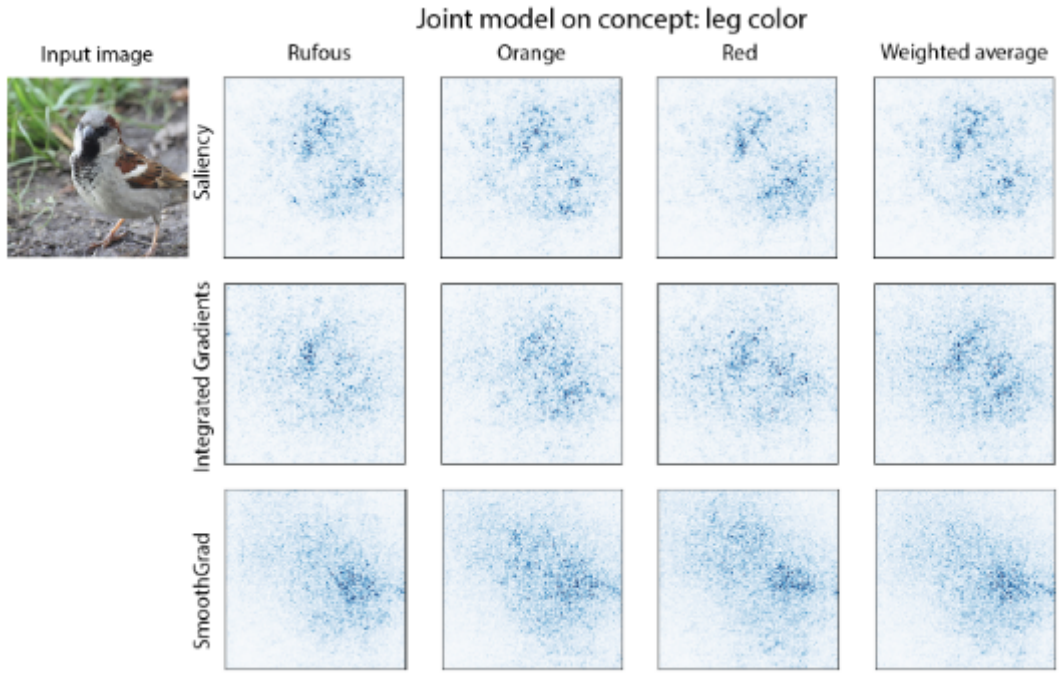
Criticism (Margeloiu et al, ICLR workshop, 2021)

Concept-based predictions seem to use information from the entire image



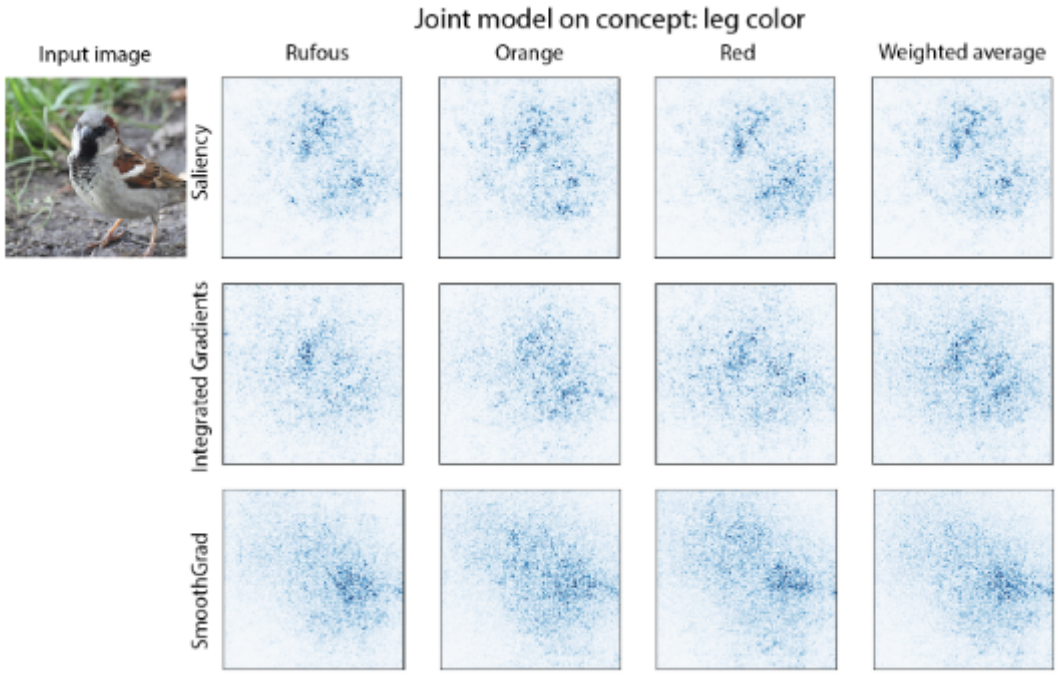
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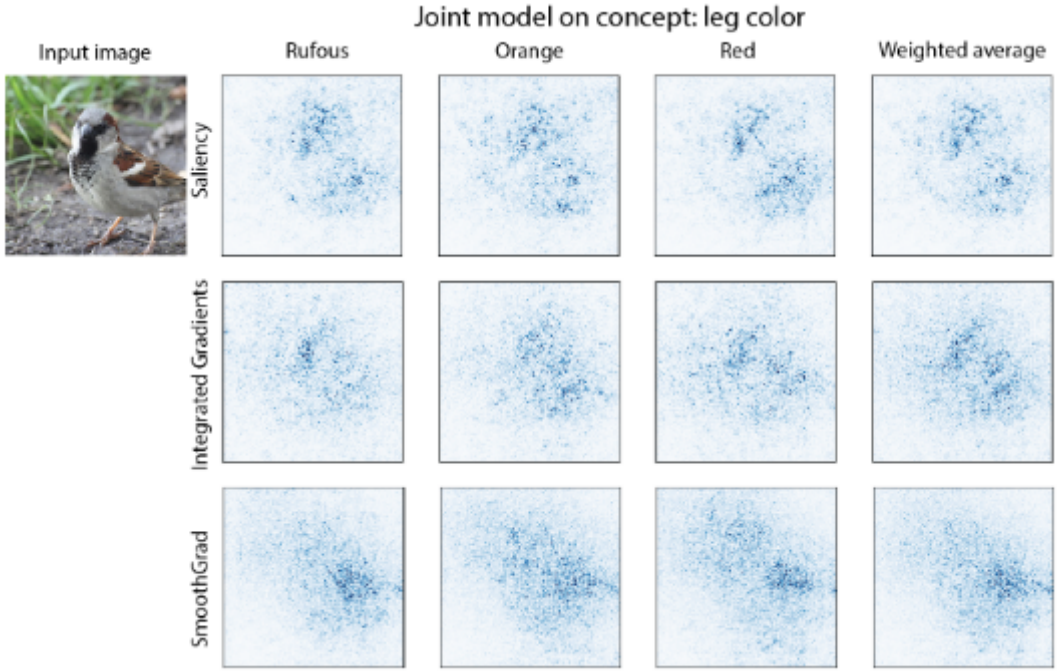
Concept-based predictions seem to use information from the entire image



Why does this happen? Is this a problem?

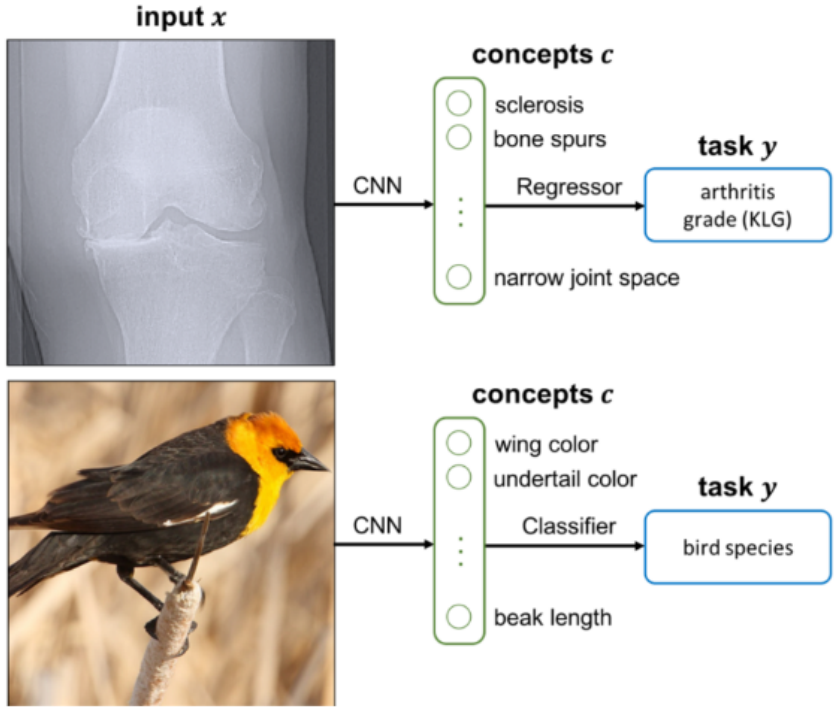
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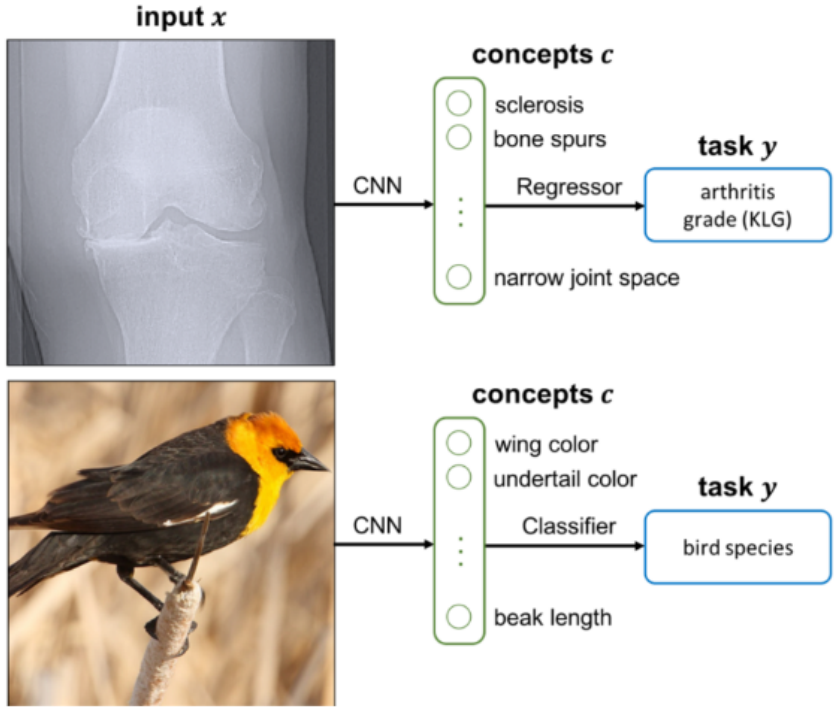
Discovered by previous students in this course:
Majority voting used as dataset preprocessing likely leads to an incorrect reasoning: Image –> species –> conceps –> species

Criticism 2 (Havasi et al, NeurIPS 2022)



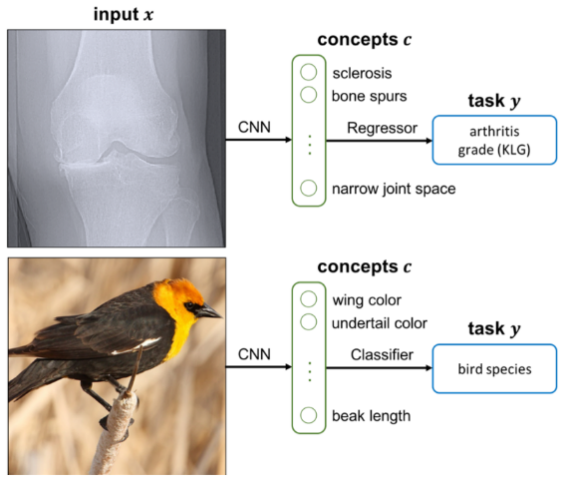
- One way to train CBMs is to train the predictive sub-networks independently using ground truth concepts, not predicted ones.
- This can lead to underperformance – why?

Criticism 2 (Havasi et al, NeurIPS 2022)



- One way to train CBMs is to train the predictive sub-networks independently using ground truth concepts, not predicted ones.
- This can lead to underperformance – why?
- **Common solution:** Train on soft, predicted concepts

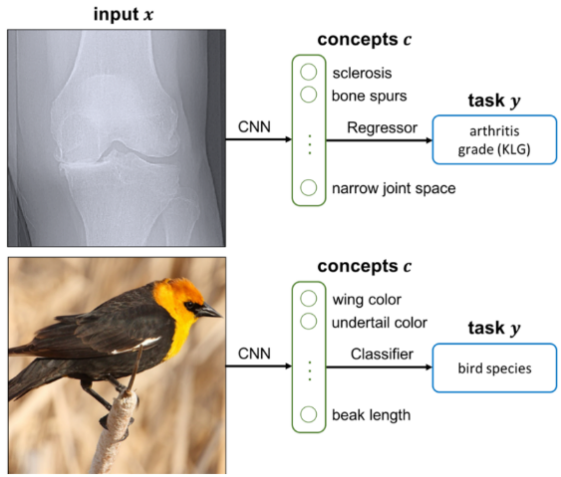
Criticism 2 (Havasi et al, NeurIPS 2022)



This paper identifies two sources of problems due to training on predicted soft concepts rather than hard annotated concepts:

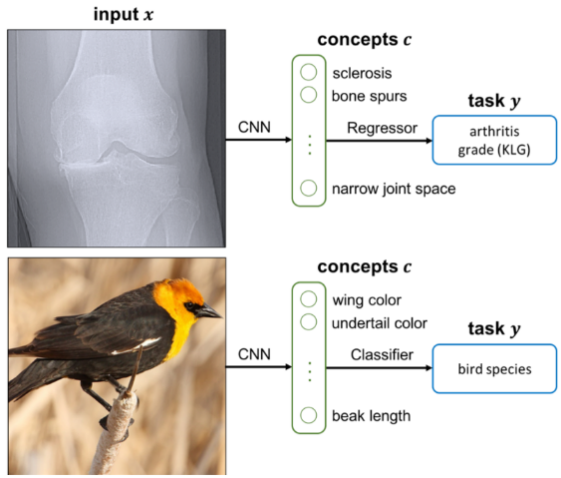
- Correlated concepts
- Leakage when soft concepts c contain more bits than discrete ones, and can store more information than you realize.

Correlated concepts



- Affects intervention: Changing one concept but not the correlated one may not have the wanted effect
- Affects interpretation: Might misjudge concept importance
- Affects performance: Model may be unable to encode e.g. mutually exclusive concepts

Leakage in CBMs



- Leakage when the Markovian assumption fails, that in $x \mapsto c \mapsto y$, y is conditionally independent of x conditioned on c – i.e. x does not contain additional information about y that is not present in c.
- Solution: A side channel is introduced to encode additional information from the input feature

Validating XAI

Summary: Intrinsic prototype-based XAI

- You have seen examples of intrinsic, prototype-based XAI
- These may have been overhyped, but are nice models when they are applicable
- Often require specialized solutions and specialized data – limited applicability

